

6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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Abstract.

Understanding climate change relies heavily on grasping the role of cloud radiative effect and its climate feedback. However, these factors pose the greatest uncertainty among atmospheric processes. This uncertainty stems partly from the limited global-scale cloud measurements. To address this gap, our study presents a six-year statistical analysis of cloud coverage and microphysics properties at the ACTRIS-CCRES AGORA observatory in Granada, Spain. Here, we utilized CLOUDNET post-processed data (e.g target classification, liquid water path, droplet effective radius and so on) of remote sensing instruments and microwave radiometer thermodynamic profiles. The analysis focuses on single layer cloud properties comparison for liquid, ice, mixed phase, liquid with rain, and mixed phase with rain clouds. Therefore, a distinct seasonal pattern in cloud occurrence was observed. During the summer (June-August), single layer clouds occurrence is less than 10%, and precipitation is absent. Conversely, cloud occurrence significantly increases during winter (December-February) and spring (March-May), where Ice and mixed phase clouds with rain are the predominant type. Additionally, a negligible presence of liquid clouds with rain were found. Seasonal trends in liquid water content (LWC) and droplet effective radius are unclear for mixed-phase and liquid clouds. Effective radius remains stable year-round, around 5 μm for liquid clouds below 3 km, while for mixed-phase clouds it can vary between 5 and 12 μm , mainly found at 1-2 km and 5-9 km. LWC shows higher values in cold periods and lower in summer, but with significant variability across both cloud types. Same seasonal patterns for ice water content (IWC) in ice and mixed-phase clouds were challenging to identify. Monthly profiles showed little variation, although values tended to rise in summer. Ice clouds typically have effective radius between 20-30 μm at 8-12 km altitude, peaking at 65 μm in summer. Mixed-phase clouds exhibit ice effective radius around 50 μm between 1-7 km altitude. The mean geometric thickness of ice and mixed-phase clouds throughout the seasons is 1.2 ± 1.3 km and 2.0 ± 1.6 km, respectively, contrasting sharply with the 0.2 ± 0.2 km observed in liquid clouds. The monthly variability in thickness is substantial enough to obscure any seasonal differences. Similarly, the analysis of cloud base height reveals mean values of 1.9 ± 2.0 km (excluding summer) for liquid clouds, 4.8 ± 2.2 km for mixed-phase clouds, and 7.0 ± 2.4 km for ice clouds. During summer a different vertical distribution of cloud base was found due to different mechanisms of cloud formation. Conclusively, a statistical analysis was conducted on single-layer cloud properties in the Southeast of Spain, highlighting the mainly cloud type associated with rainfall in the region.

1 Results and discussion

1.1 Classification method comparison

A comparative analysis between the two classification algorithms were conducted in order to evaluated their applicability for further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories generated by these methods. The findings from this comparison will inform the selection of the most appropriate method for future atmospheric studies and cloud analysis. In this framework, Figure 1 shows a case study comparing cloud classification by time and classification by hydrometeor grouping for an mixed phase cloud, where red and black dots represent the cloud base and top, respectively. As expected, the classification by time lead to the fragmentation of clouds into different types. In this case, the same cloud are being classified as ice and mixed-phase by time. On the other hand, the right panel presents the classification by grouping, which preserves the homogeneity of clouds, providing a more realistic classification.

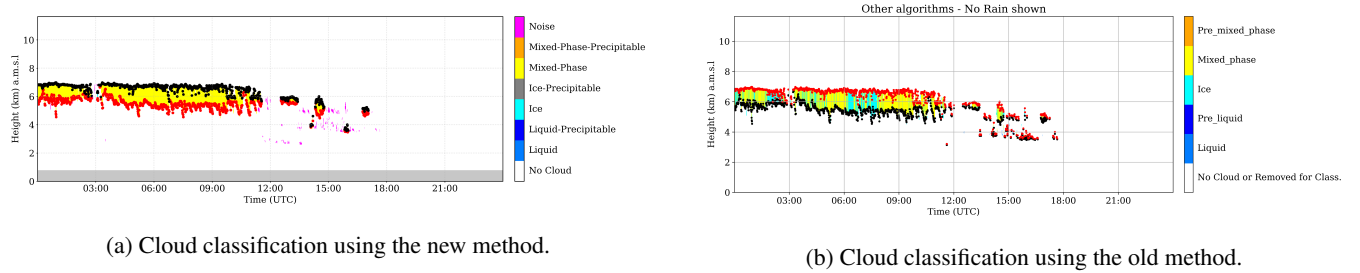


Figure 1. Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is done by time.

As a consequence, the traditional method (by time) of cloud classification can generate high correlations between cloud categories in terms of cloud properties and frequency of occurrence. For example, right panel of Figure 1 show that both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, it is clear that this occurs only because of the method of classification. Another example on how can the method influence in the cloud properties are shown in figure 2. This figure shows the daily average of IWP for ice clouds (bellow panel) and mixed-phase clouds (above panel) using the classification by time. The closeness daily ice amount between these two clouds and the clear similar pattern suggest a correlation generated by the cloud classification. It is worthy to note that this section is not focused on the analysis of the IWP, this will be done in section XXX. Instead of this, here we centred on the correlation generated by the classification algorithms.

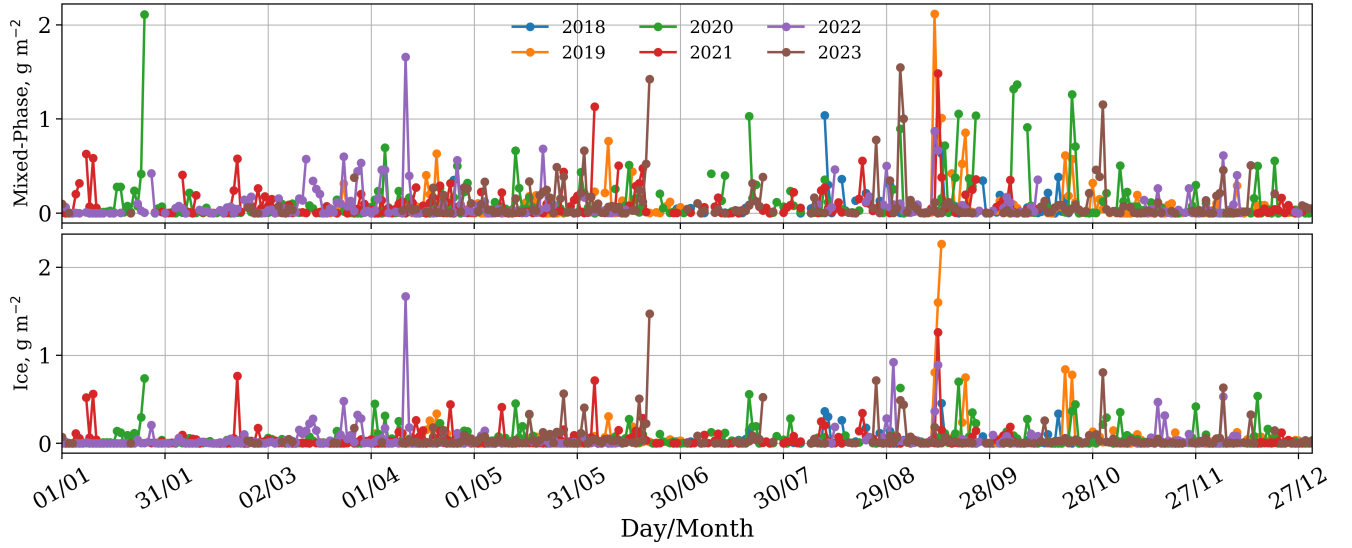


Figure 2. Time series of Mixed-Phase (top) and Ice (bottom) cloud water content from 2018 to 2023. Each year is represented by a different colored line, illustrating seasonal variations and interannual variability in the concentrations of mixed-phase and ice cloud water content.

Therefore, in order quantify the correlation effect generated by both algorithm, a correlation analysis on daily cloud occurrence amid all cloud categories were performed. Figure 3 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the classification by hydrometeor grouping on the left, and for the classification by time on the right. As can be seen, the new proposed method demonstrate generally weaker correlations among cloud types, with most values clustering closer to zero. For example, the highest correlations observed were 0.19 between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, such as between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In addition, computed cloud properties will be not dependent on the method as they must be, avoiding some misinterpretation of the results.

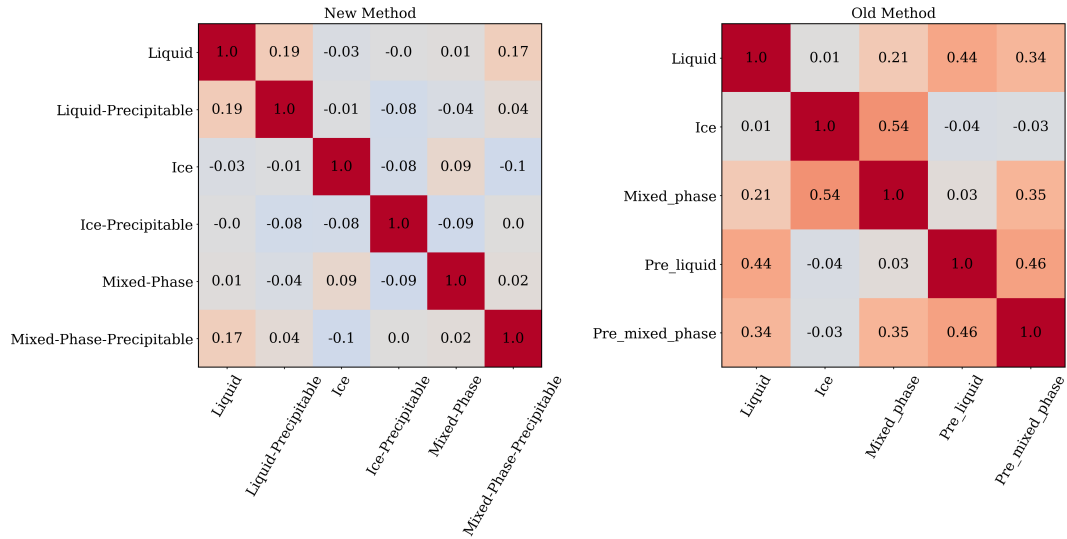


Figure 3. Correlation matrices comparing the daily occurrence between different cloud types using two methods. In the left, the New Method exhibits generally weaker correlations, indicating clearer distinctions between cloud types. In the right, The Old Method shows stronger correlations, suggesting more pronounced interdependencies and potential overlaps between cloud types

In contrast, the classification by time exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed_phase clouds was 0.54, and between Liquid and Mixed_phase clouds was 0.44, highlighting significant interdependencies between these cloud types. Furthermore, the old method showed higher correlations between Liquid and Pre_liquid (0.34), Ice and Pre_mixed_phase (0.35), and Pre_liquid and Pre_mixed_phase (0.46), indicating a less distinct separation between cloud types. These higher correlation values are a consequence of the classification scheme defined in this method. The cloud classification by time can fragment them, allowing tagging cloud profiles in different cloud categories, while they still inside the same cloud structure.

Overall, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing between different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

1.2 Daily cycle for single layer clouds

Using the classification algorithm by hydrometeor grouping, we investigate the diurnal cycle of cloudiness, with a particular focus on seasonal variations. By analyzing the patterns of cloud occurrence, we aim to uncover the influence of local convection and synoptic conditions. Our research delves into the diurnal behavior of convective clouds, particularly during spring when solar incidence and temperature variations are pronounced. This analysis not only highlights the critical periods of cloud and rain occurrence but also elucidates the potential drivers behind these patterns, offering valuable insights into regional climate dynamics.

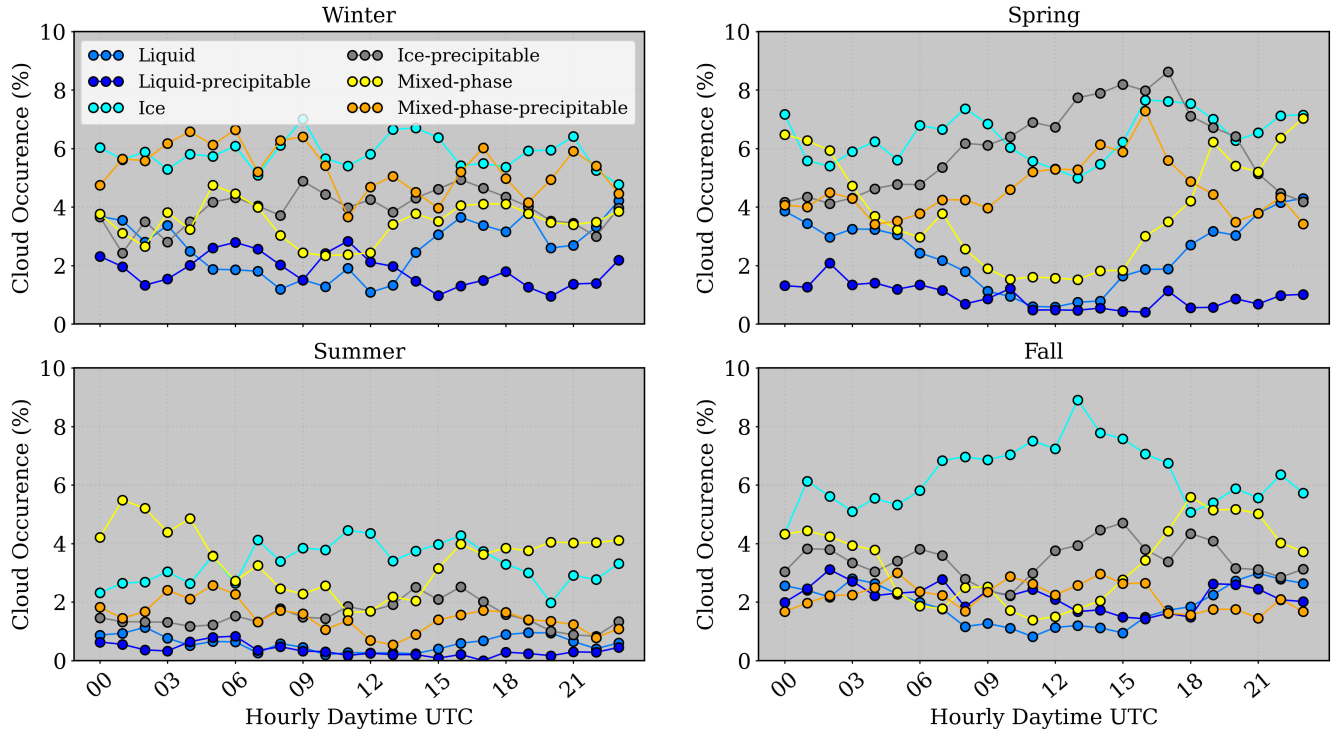


Figure 4. Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

Figure 4 shows the possible presence of convective clouds during spring, with a notable occurrence of precipitable ice clouds observed between 9 and 16 hours, coinciding with high solar incidence. This behaviour, which begins in winter, becomes more pronounced in spring as temperatures rise. Our results indicate that cloud occurrence and rain events are generally high in spring. These observations are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

1.3 Single layer cloud properties evaluation

1.4 Cloud occurrence and microphysics properties

By converting hydrometeors into cloud types using the cloud classification algorithm (refer to section ??), monthly frequency of single-layer clouds occurrence was obtained as depicted in Figure 5. Indeed, all subsequent analyses focus on single-layer clouds, as the CLOUDNET target classification product can be highly inaccurate in multilayer cases due to errors in LWP partitioning (?). In this scenario, this figure shows a predominant presence of precipitating mixed-phase clouds (Indicated as

Pre-Mixed-Phase). Alternatively, almost no precipitation due to liquid clouds was observed, just a slight maximum of 5% in the transition period between fall and winter. Next, ice clouds are the second most present cloud type, followed by mixed-phase clouds and liquid clouds. However, the presence of precipitating mixed-phase clouds are significantly higher than other cloud types, with a substantially stronger seasonal pattern. Its maximum frequency coincides with the cold regimes, in winter and spring, where their frequency reaches 25% and 30%, respectively. Evidently, the minimum occurs in summer (June - August) for all cloud types, when temperature is high and relative humidity is much lower. Also, during summer ice clouds prevailed and no significant liquid precipitation was registered.

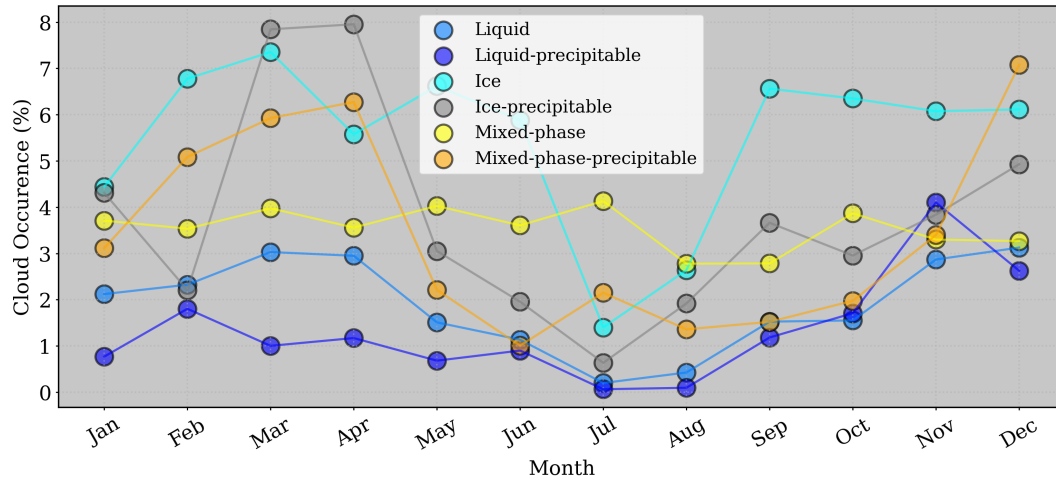


Figure 5. Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

Next, seasonal evolution of cloud thickness (ΔZ), base and top heights are displayed in Figure 6. In relation to cloud thickness, mixed-phase and precipitable mixed phase clouds exhibit the largest and most variable values, followed by ice, precipitating liquid clouds and liquid clouds. A notably broader distribution for precipitable mixed-phase clouds were observed in summer, with a median of 3 [0.6 to 5.1] km. Where values in brackets represents the 25th and 75th percentiles, which demonstrate a greater uncertainty for this variable. Moreover, in this season, these clouds have less than 4% of occurrence (refers to figure 5). Throughout the remainder seasons, median cloud thickness for precipitable mixed-phase clouds are raging from 1 to 1.4 km, close to non-precipitable clouds thickness, ranging from 1.4 to 1.7 km during the entire period. The significant high values Annually, Ice cloud thickness range from 0.7 to 0.9 km, precipitable liquid clouds from 0.27 to 0.8 km, and liquid clouds, from 0.18 to 0.2 km. In general, cloud-thickness distribution is skewed toward large thickness throughout all seasons, and the mode is consistently smaller than the median. Additionally, there is no evident seasonal pattern for this parameter due to its considerable variability.

Subsequently, the middle panel of figure 6 shows that ice clouds have the highest cloud base height, followed by mixed-phase, precipitable mixed phase, and both liquid clouds, precipitable and non-precipitable. In summer, cloud base distribution experiences the most pronounced alterations. For instance, liquid clouds exhibited a wider and higher spread distribution, with

a median height of 3 [1.8 to 4.9] km. In contrast, their median ranged from 1.4 - 1.8 km for the remaining period. In relation to mixed-phase clouds, its base height distribution becomes sharper in summer, with a median height of 5.2 [4.6 to 6.0] km, practically the same of fall, at 5.4 [4.0 to 7] km. For the rest of the seasons, their median height is approximately 1 km lower, between 3.7 and 4.0 km. Moreover, precipitable mixed-phase cloud bases are significantly higher in summer, with a median height of 3 [1.8 to 4.9] km, in contrast with the remaining seasons, varying between 1.2 - 1.9 km. Precipitable liquid clouds also presented a higher and broader cloud base in summer, with a median height of 1.7 [1.8 to 4.9] km, about 1 km higher than other seasons, which fluctuates between 0.7 to 1.2 km. For ice clouds, despite exhibiting a higher median cloud base in fall, during summer, an uncommon mode in comparison with other seasons is observed at 5 km - indeed, this is also observed for liquid and mixed-phase clouds. During fall, ice cloud base height exhibits a median value of 8.1 km [6.7 to 9.3] km, also 1 km higher than other seasons, quite well stable in 7.0 km.

Bottom panel of Figure 6 shows a similar seasonal pattern for cloud top height, where the main difference was observed in summer. Particularly, for precipitable mixed-phase clouds, cloud top height has substantially greater variation than cloud base height, with skewed distributions toward large heights. Evidently, this variability reflects in cloud thickness as previously observed. Thus, cloud thickness is directly affected by cloud top measurements in precipitable clouds. Since the CDR reflectivity experienced greater attenuation in this case, cloud top height could be underestimated in some cases, leading to the observed variations. For these clouds, median cloud height, normally encountered between 2.3 - 3.7 km, were found in 6.2 [3.4 to 8.4] km in summer. Similarly, precipitable liquid clouds which have a median top height around 1.4 - 1.7 km, reached at 2.2 [2.0 to 2.6] km in summer. Alternatively, ice and mixed-phase clouds reached a maximum median height in fall of 9.5 [8.3 to 10.4] km and 8.2 [6.3 to 9.6] km, respectively. For the rest of the period, their median top height stayed between 8.5 - 8.7 km and 6.3 - 6.7, respectively. Subsequently, liquid clouds presented larger top height variability in summer, similar to its cloud base height distribution. During this particular season, their median top height is 2.9 [1.8 to 4.9] km, roughly 1 km higher than other seasons, which are varying from 1.4 - 1.8 km.

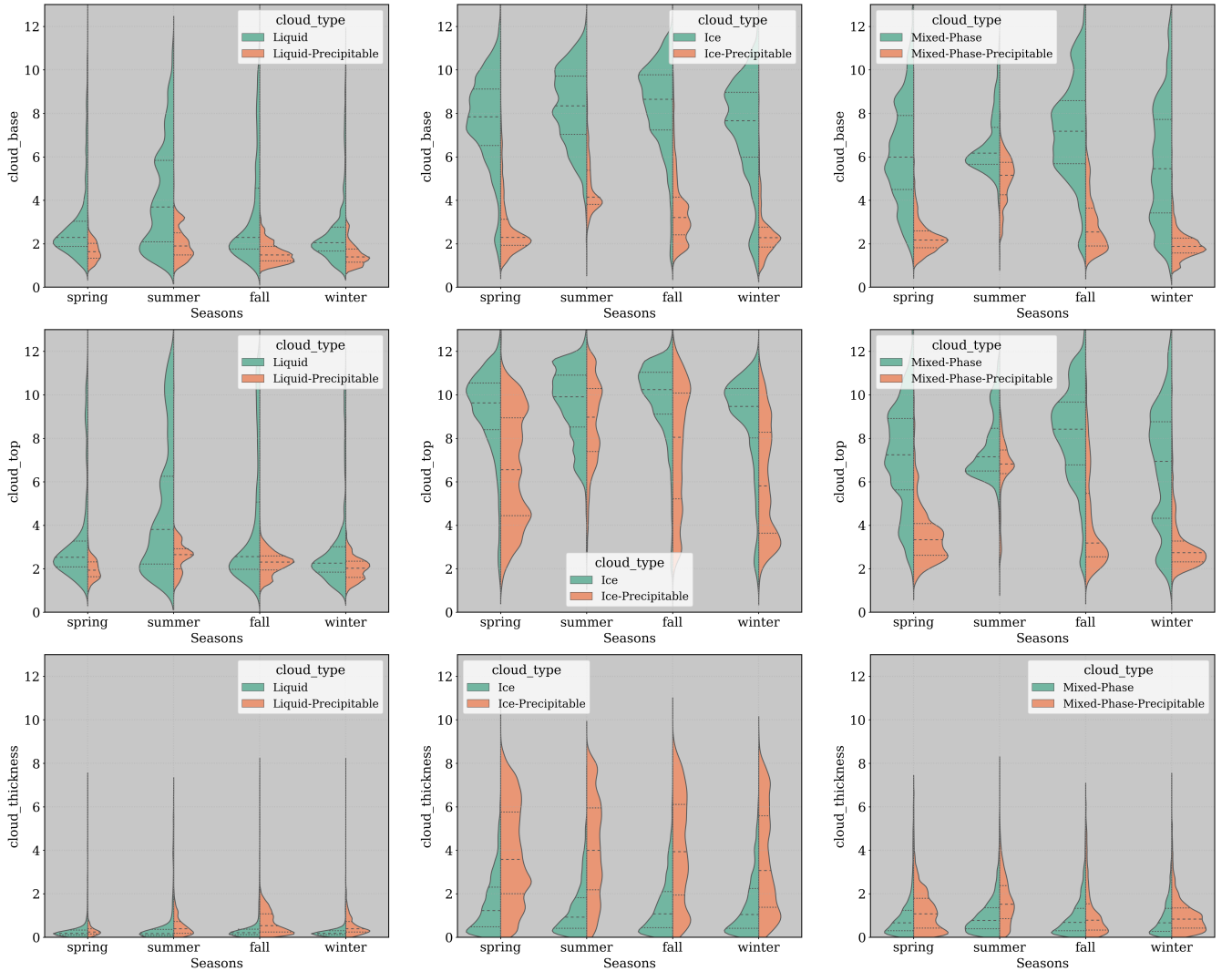


Figure 6. Seasonal evaluation of cloud thickness, base and top. The legend above indicates the color correspondent to each cloud type. This figure shows a dark gray box-plot with a white median inside an estimated density distribution. This distribution is displayed in relation to the middle of the box-plot in the right and the left direction.

130 In general, cloud base and top height is higher in summer for almost all cloud types, except for ice and mixed-phase clouds, which reach their peak during fall. Despite that these properties showed a seasonality in terms of median values, the interquartile range showed strong variability, especially for ice and mixed-phase clouds. Specifically, precipitable mixed-phase clouds exhibited heightened variability in cloud top height, notably less precise compared to their base height. This suggests a possible underestimation of cloud tops due to radar reflectivity attenuation by rain droplets. Same behavior would be observed

135 in precipitable liquid clouds, but they are normally thinner and easier to be totally penetrated by the radar signal. Also, these clouds do not have enough occurrence to strengthen the statistics of their macrophysical properties, as can be seen in Figure 5.

1.5 Liquid water content and liquid water path

Monthly and seasonal evolution of liquid water content (LWC) and liquid water path (LWP) for liquid clouds are displayed in Figure 7. Left panel shows the mean seasonal profiles of LWC, and the right panel shows the monthly mean profiles of LWC and monthly mean values of LWP (right-bottom panel). Since the monthly median of LWC is usually zero for all cloud types, suggesting a dry atmosphere with a typical water shortage, we decided to analyze average values. First, mean seasonal profiles showed peaks of LWC about 0.07 g m^{-3} at 1 km, 0.1 g m^{-3} at 1.1 km, and 0.14 g m^{-3} at 1.7 km during fall, winter and spring, respectively. In spring, the average value at 4 km was strongly affected by the presence of outliers, identified in May by the monthly distribution on the right panel (note the dark red tone), creating an abnormal peak at this level. Despite that summer showed a sharp peak of about 0.16 g m^{-3} at 200 m, above this level, this season has lower average LWC in comparison with other seasons. In addition, the monthly evaluation of LWC, displayed in the right panel, points out its tremendous variability, which passes through several orders of magnitude. This panel shows that relevant LWC are below 2 km, in addition its average decreases two orders of magnitude in July and August (during summer) but are well vertically distributed, reaching 5 - 6 km. This spatial arrangement is in agreement with the cloud base and top shown in Figure 6. Also, the presence of small values occur much more frequently than large values, leading to several outliers which are affecting the mean LWP, as can be seen in the bottom-right panel of figure 7. The mean LWP is denoted by the black star, and the median by the box-plot red line. A large discrepancy between these two statistical parameters was observed, except in summer where values are all small and there are relatively low cloud liquid profiles, as indicated by the blue lines. In general, average LWP stays below 100 g m^{-2} for liquid clouds, with 78% of frequency of occurrence and exhibit a clear seasonal pattern, with minimum in summer and maximum in winter-spring. On the other hand, the median does not exhibit a significant seasonal pattern.

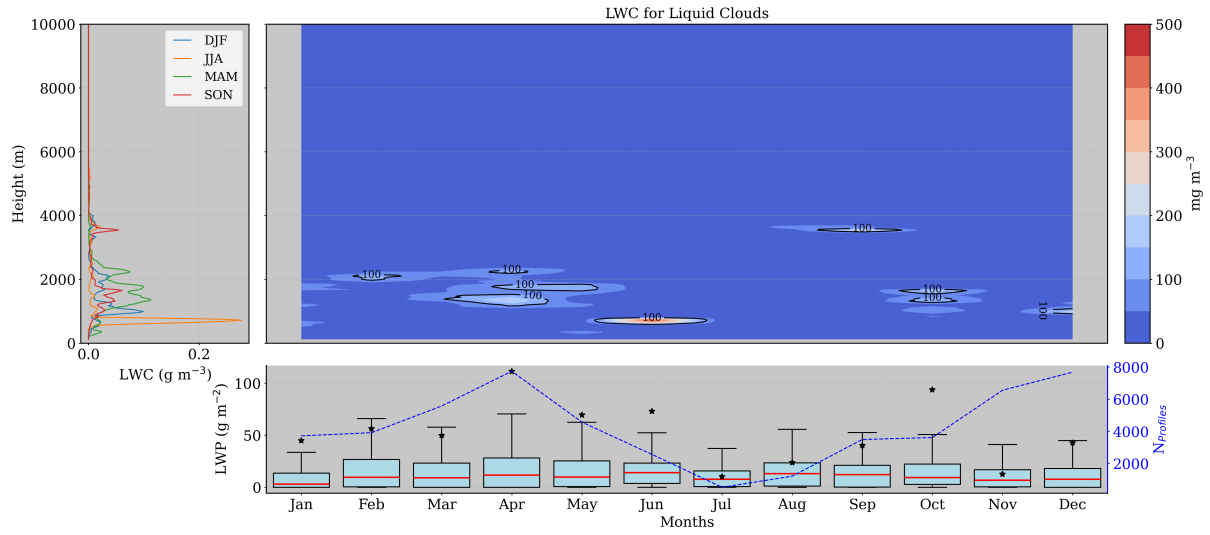


Figure 7. Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal mean profiles of LWC. Right panel shows the monthly mean profiles of LWC and monthly box-plot of LWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Similar seasonality was observed for precipitable liquid clouds, depicted in Figure 8. Nevertheless, LWC are higher due to the presence of drizzle, and peaks are more sharply defined around 1 km during fall, winter and spring. Also, winter achieved higher values, twice as large as in fall and spring (0.2 g m^{-3}), reaching 0.7 g m^{-3} . In summer, values are relatively smaller, with peak of 0.1 g m^{-3} around 2 km. Right panel shows that monthly values still fluctuate through several orders of magnitude, and larger values are not well vertically distributed in summer as observed for the only liquid case. They still kept within thin layers, although they reached at slight higher levels. This is in agreement with vertical position of cloud base and top height (see Figura 6), which does not showed a significant vertical variability in summer. In terms of LWP, no seasonality was observed due to the presence of rain droplets, which highly increased its water content in all months. Note that LWP were not separated between cloud and rain, it is the integrated value reported by the microwave radiometer. As a result, slight discrepancy between the mean and the median was observed, and its values are roughly 10 - 20 times higher in comparison with the liquid case, with 75% of data below 250 g m^{-2} . Also, October presented a significant variation of LWP in relation to other months, with more than 1000 g m^{-2} inside the 75th percentile. Thus, this month is marked by stronger rains from this particular cloud.

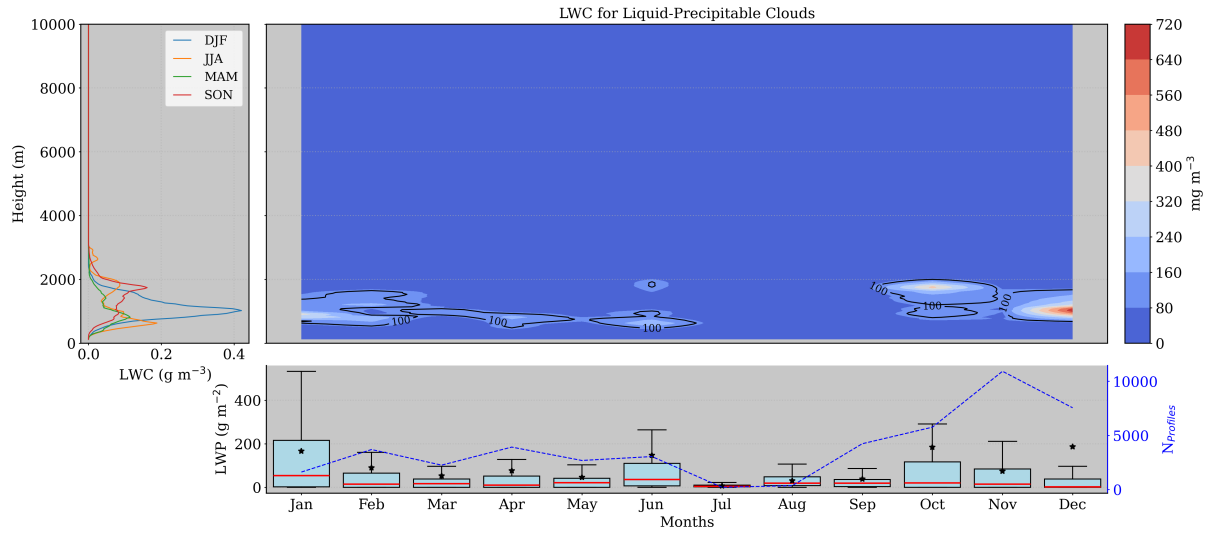


Figure 8. Same as Figure 7, but for precipitable liquid clouds.

Mixed-phase clouds exhibited marginally greater occurrence than liquid clouds (Figure 5). Typically, they co-occur and can be comparable in terms of water content. Seasonal average profiles of LWC for mixed-phase clouds in the left panel of Figure 9 shows a similar vertical distribution of LWC between summer and fall (warm regimes), and between winter and spring (cold regimes). Below 4 km, cold regimes have higher average LWC, and similar values with warm regimes above it. In general, these profiles have just smaller differences, and average values are always less than 0.05 g m^{-3} . This shows that liquid clouds, besides being thinner, have more water than mixed-phase clouds, suggesting that mixed-phase clouds are mainly composed by ice hydrometeors. Next, monthly mean time evolution on the right panel shows a spreader vertical distribution of LWC, which becomes thinner and higher in summer. This suggests that liquid layers inside mixed phase clouds become thinner in summer, and the ice content is still high enough to keep its thickness around the seasons, as observed in figure 6. Additionally, the right-bottom panel of Figure 9 shows greater occurrence of large LWP values at the end of winter and during spring. Overall, 84% of the analyzed mixed-phase cloud profiles have LWP below 100 g m^{-2} , emphasizing the point these clouds have less water than liquid clouds, which has 78% as already discussed.

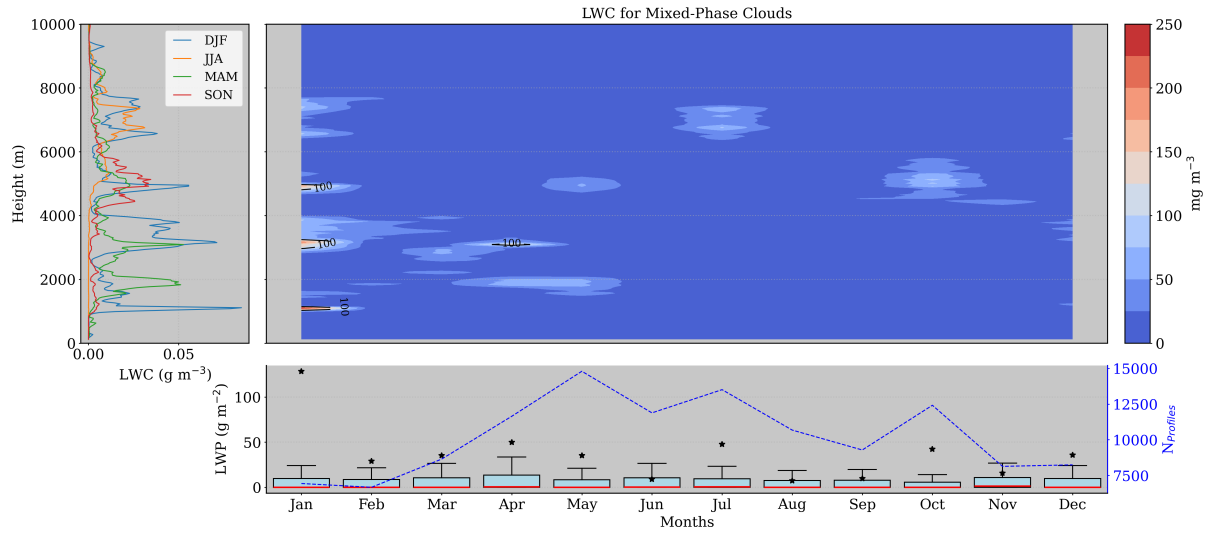


Figure 9. Same as Figure 7, but for mixed-phase clouds.

Left panel of Figure 11 reveals that precipitable mixed-phase clouds present similar average profiles up to 4 km in fall, winter and spring. During summer, average LWC within this layer are more than twice smaller in comparison with other seasons. On average, these clouds still have less water content than precipitable liquid clouds below 2 km. This evidence, combined with the observation that liquid layers are notably narrower than mixed-phase layers, suggests that liquid hydrometeors represent a mere fraction of mixed-phase clouds. This observation has more weight here since precipitable mixed-phase clouds have much more frequency of occurrence. In addition, the blue axis in the bottom-right panel of Figure 11 shows a relevant higher number of cloud profiles, with a sudden decrease in summer. Monthly evolution of LWC clearly indicates a concentration of LWC below 4 km, except in summer, when these clouds have a higher cloud base. In general, the observed mean values are higher than only mixed-phase clouds due to the elevated presence of drizzle. In this case, LWC has a narrower vertical distribution and is closer to the ground. Therefore, they will also have much higher LWP than mixed-phase clouds, as depicted in the right-bottom panel. In terms of its monthly median, no clear seasonal pattern is identified. However, as in the precipitable liquid case, a strong variability of LWP was found in October. This time, we can assume with more statistical confidence that rain events in October are more intense. Although precipitation is more frequent during spring. Moreover, precipitable mixed-phase clouds exhibit 70% of LWP below 250 g m^{-2} ; in other words, they can attain higher values than precipitable liquid clouds.

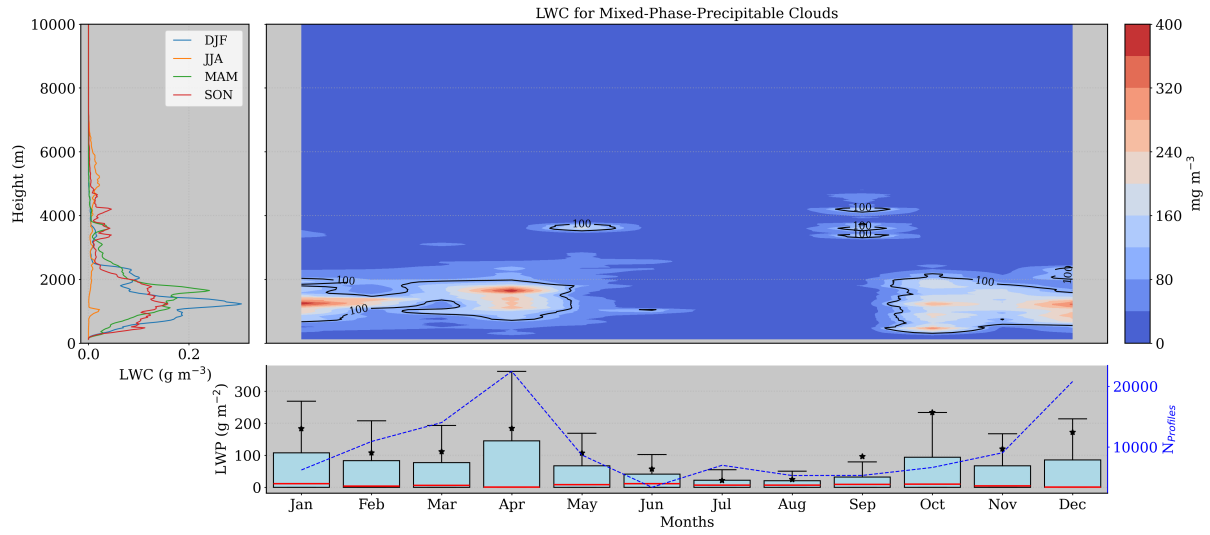


Figure 10. Same as Figure 7, but for precipitable mixed-phase clouds.

Overall, a notable reduction on LWC was observed in summer. Also, values were found at high levels in the atmosphere in this season, in accordance with the cloud base height pattern. These differences were more readily observed by mixed-phase and precipitable mixed-phase clouds. On the other hand, no seasonal pattern of LWP was clearly identified for all clouds, owing to the high month variability of this variable. To conclude, no clear seasonal pattern was identified for clouds in relation to monthly median LWP. However, liquid clouds have greater LWP than mixed-phase clouds. Once mixed-phase clouds are considerably thicker than liquids, the majority of their composition comprises ice hydrometeors, which will be confirmed in the next section. Moreover, for precipitation cases, the liquid water content increases due the presence of drizzle, and LWP can reach values approximately 10 times greater. In addition, precipitable mixed-phase clouds can achieve higher values of LWP than precipitable liquid clouds.

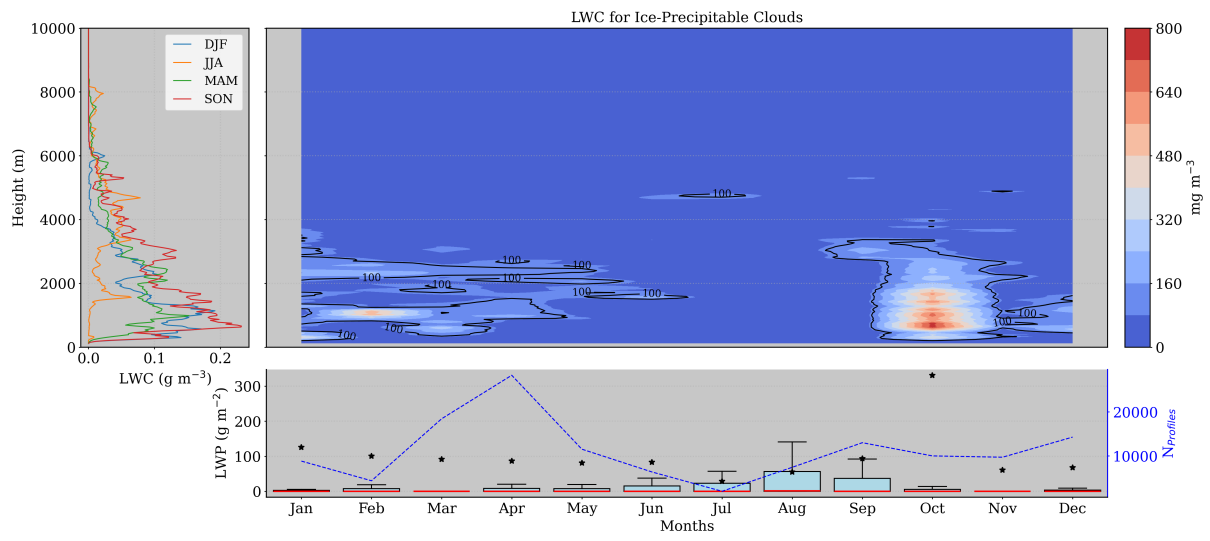


Figure 11. Same as Figure 7, but for precipitable ice clouds

1.6 Ice water content and ice water path

Figure 12 displays the average IWC content profiles for each month and seasons, and monthly box-plots of IWP. An evaluation of seasonal profiles on the left panel of this figure denoted a markable presence of IWC at 2 km in winter, with a peak of 0.2 g m^{-3} , while at the same level, no IWC was observed during other seasons. Therefore, despite that Figure 6 shows almost no ice cloud base at this level during winter, the presence of just a few clouds with high ice water content are affecting its average. In fact, monthly profiles in the right-top panel reveal that this ice cloud occurs in January. This can be seen by the red tone centered exactly at 2 km in this month. During spring, the peak is around 0.12 g m^{-3} at 4.5 km, and above this level its average profile coincides with the winter profile. A similar feature is observed for warm regimes, average profiles of fall and summer coincide above 7 km. Below this level, fall has a peak of 0.2 g m^{-3} for about 5 km and summer has a peak of 0.1 g m^{-3} around 6 km. In general, ice cloud base heights are usually above 6 km (refers to Figure 6). Hence, relevant values of IWC for these clouds are within 6 and 10 km, in addition, there is no IWC below 2 km, except in January. Concerning the IWP, the median are relatively smaller than the mean, showing a right-skewed distribution with outliers, which are affecting their monthly mean. In terms of medians, no clear seasonal pattern was found. However, larger variability was observed during spring and beginning of fall. In fall (Sep-Nov), ice cloud bases are higher, and average IWP also experienced a maximum of about 100 g m^{-2} in September. Regarding the entire dataset, 90% of the IWP are smaller than 100 g m^{-2} .

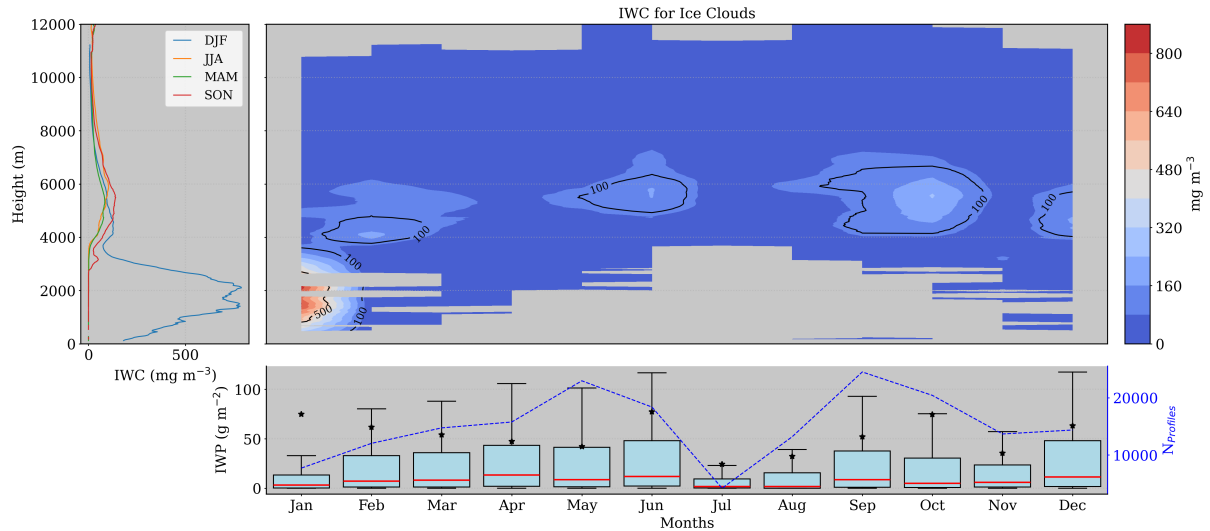


Figure 12. Seasonal and monthly average of ice water content (LWC) and ice water path (LWP) for ice clouds. Left panel shows the seasonal mean profiles of IWC. Right panel shows the monthly mean profiles of IWC and monthly box-plot of IWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Mixed-phase clouds exhibited a similar mean profile of IWC during summer and fall as can be seen in the left panel of Figure 13. The only significant difference was during the summer, when mixed-phase clouds do not have a peak around 2 km as observed for ice clouds. Although seasonal profiles suggest mixed-phase clouds have almost the same amount of IWC of

ice clouds, it is worth noting that cloud thickness for ice clouds is smaller than for mixed phase clouds. It was observed that mixed phase clouds are almost 1 km thicker than ice clouds. Thus, IWC will be greater, which is the integrated IWC, will be greater as can be seen by the box-plots in the right-bottom panel. This panel shows that IWC easily surpasses 50 g m^{-2} , which was not common for ice clouds. In addition, there is less discrepancy between monthly mean and median than for the ice case.

225 Also, 74% of all IWP for mixed-phase clouds are below 100 g m^{-2} , showing that these clouds have more ice content than ice clouds.

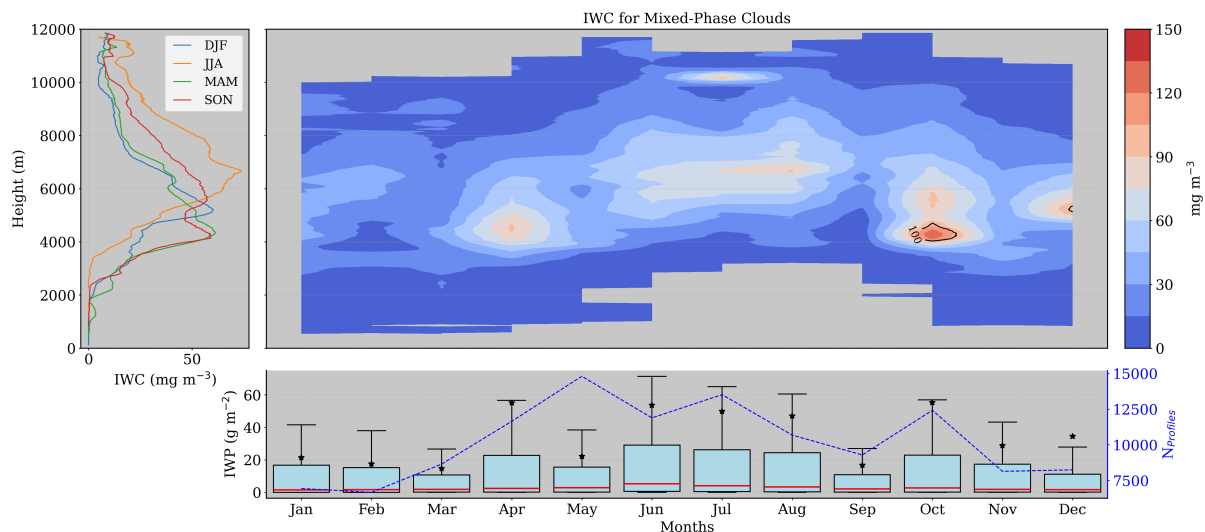


Figure 13. Same as Figure 12, but for mixed-phase clouds.

For precipitable mixed-phase clouds, winter and spring still have a similar mean profile of IWC (see Figure 14), with peaks around 0.2 g m^{-2} at 4 km. Fall still with a peak around 5 km as observed for the ice and non-precipitable mixed-phase clouds, with a just slightly higher value. The most considerable changes happened in summer, where the average profile reached values higher than the other seasons around 6 km, and kept higher up to 12 km. Within this layer, the right panel indicates larger values during summer (Jun-Aug), in red tone. As a result, IWP in this season is significantly higher in comparison with the rest of the period (see right-bottom panel of Figure 14). Although precipitable clouds have much less frequency of occurrence in summer, this frequency is just 4% smaller than mixed-phase clouds. Also, precipitable mixed-phase clouds exhibited a larger cloud thickness, almost 2 km greater than just mixed-phase clouds in this season. By comparing cloud thickness based on similar

230 higher than the other seasons around 6 km, and kept higher up to 12 km. Within this layer, the right panel indicates larger values during summer (Jun-Aug), in red tone. As a result, IWP in this season is significantly higher in comparison with the rest of the period (see right-bottom panel of Figure 14). Although precipitable clouds have much less frequency of occurrence in summer, this frequency is just 4% smaller than mixed-phase clouds. Also, precipitable mixed-phase clouds exhibited a larger cloud thickness, almost 2 km greater than just mixed-phase clouds in this season. By comparing cloud thickness based on similar

235 cloud frequencies, we can infer that precipitable mixed-phase clouds exhibit significantly greater thickness and substantially larger IWP, particularly during the summer and September. Thus, despite the fact these clouds have extremely low frequency in summer, they demonstrate greater vertical development and contain significantly more ice compared to other periods

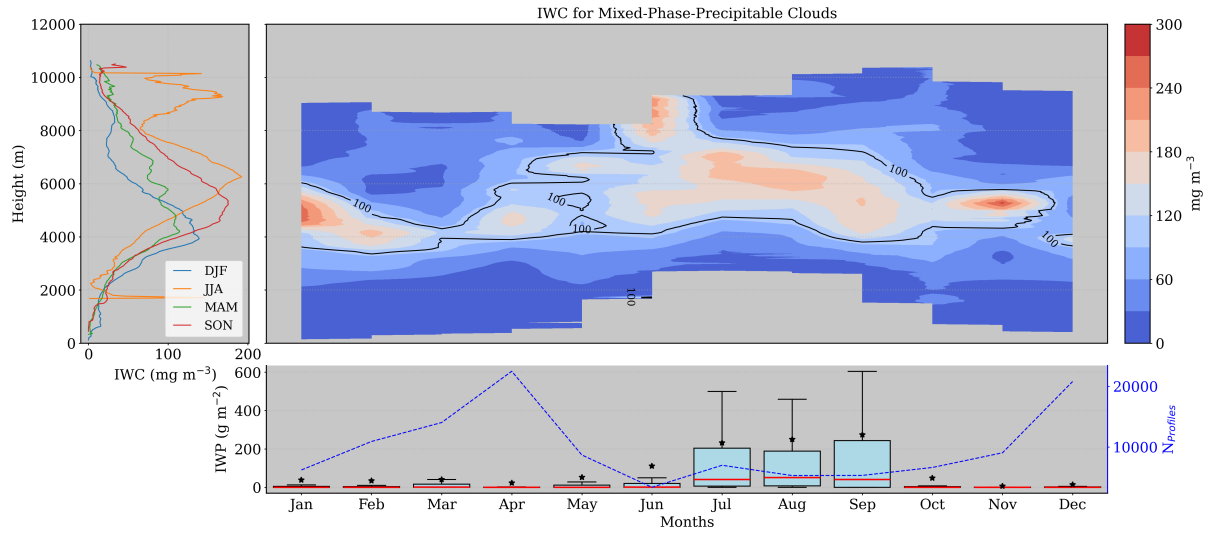


Figure 14. Same as Figure 12, but for precipitable mixed-phase clouds.

To summarize, mixed-phase clouds presented higher IWC than ice clouds. The reason can be attributed to the large thickness of the mixed-phase case compared to the only ice case, which is approximately twice as thicker. Thus, less than half of mixed-phase clouds are made of liquid hydrometers, and the rest are ice hydrometeors. Also, precipitable mixed-phase clouds showed even higher contents of ice, especially in summer. Beside their poor frequency of occurrence in summer, these clouds boast significantly greater thickness compared to mixed-phase clouds, carrying substantially more ice.

1.7 Droplet effective radius

For liquid clouds, droplet effective radius (r_e) were also analyzed by season and monthly. The following analysis was made considering only the median of r_e and not average values. Left panel of Figure 15 shows the seasonal vertical distribution of the median droplet effective radius where the shaded area represents the 25th and 75th percentiles. Right-top panel shows the monthly vertical distribution of the median, and the right-bottom panel shows the median and an estimation of the probability density function (p.d.f) of r_e by month using violin plots. These estimations were made through kernel density method and the p.d.f are shown with its reflection with respect to the middle point in the x-axis, with a kind of violin shape. Additionally, this panel has a second y axis with the number of profiles at each month to compute statistics. First, no seasonal difference was observed for profiles on the left panel of Figure 15. Second, monthly profiles on the right-top panel show that large droplets are found above 4 km, in yellow and red tones. However, monthly examination r_e distribution in the bottom panel demonstrates that the medians are about to 5 μm , corresponding to the soft blue tone below 3 km of height, consistent with the LWC distribution in Figure 7.

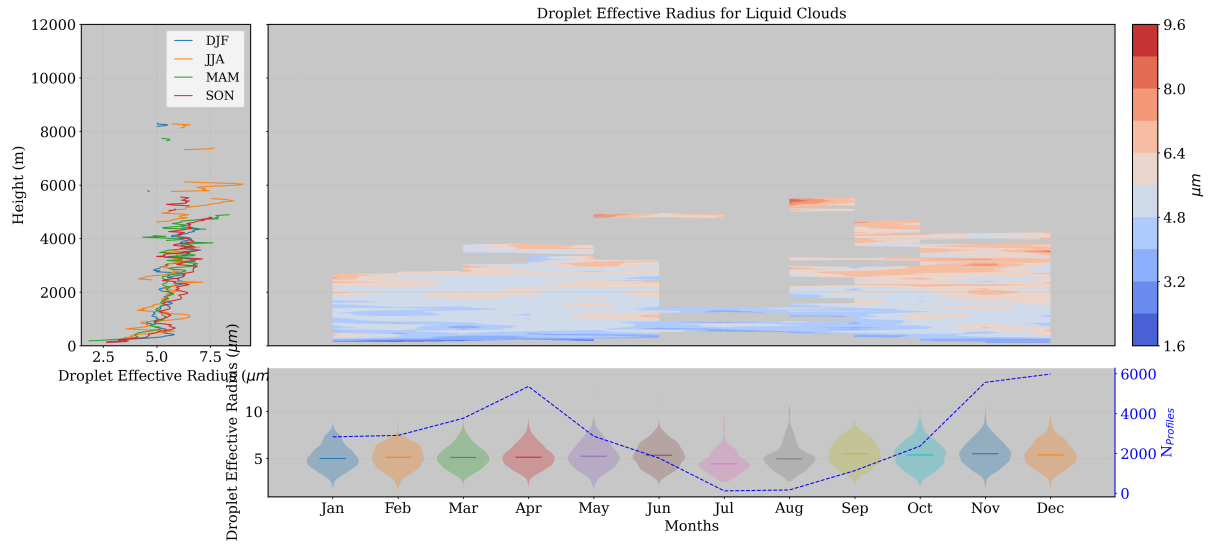


Figure 15. Seasonal and monthly statistics on droplet effective radius (r_e) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius, where shaded area denote the 25th and 75th percentile. Top-right panel shows the monthly profiles of the median r_e , while the bottom-panel displays the size distribution of r_e by months.

255 Next, precipitation liquid clouds have a slight difference on seasonal profiles below 2 km, as can be seen on the left panel of Figure 16. Below this level, spring has the lowest values, followed by winter, fall and summer. However, in this case, there are almost no profiles with precipitable liquid clouds in summer. This can be seen by the blue line in the right-bottom panel. Thus it is challenging to evaluate seasonal differences on liquid droplets for these clouds which have almost zero frequency of occurrence in summer. On the other hand, it is clear that their liquid droplets are greater than for liquid clouds. This can be associated with the presence of drizzle, as discussed in terms of LWC. In this case, the median droplet effective radius ranges around 8 and 9 μm , except in summer, where droplets are greater than 15 μm .

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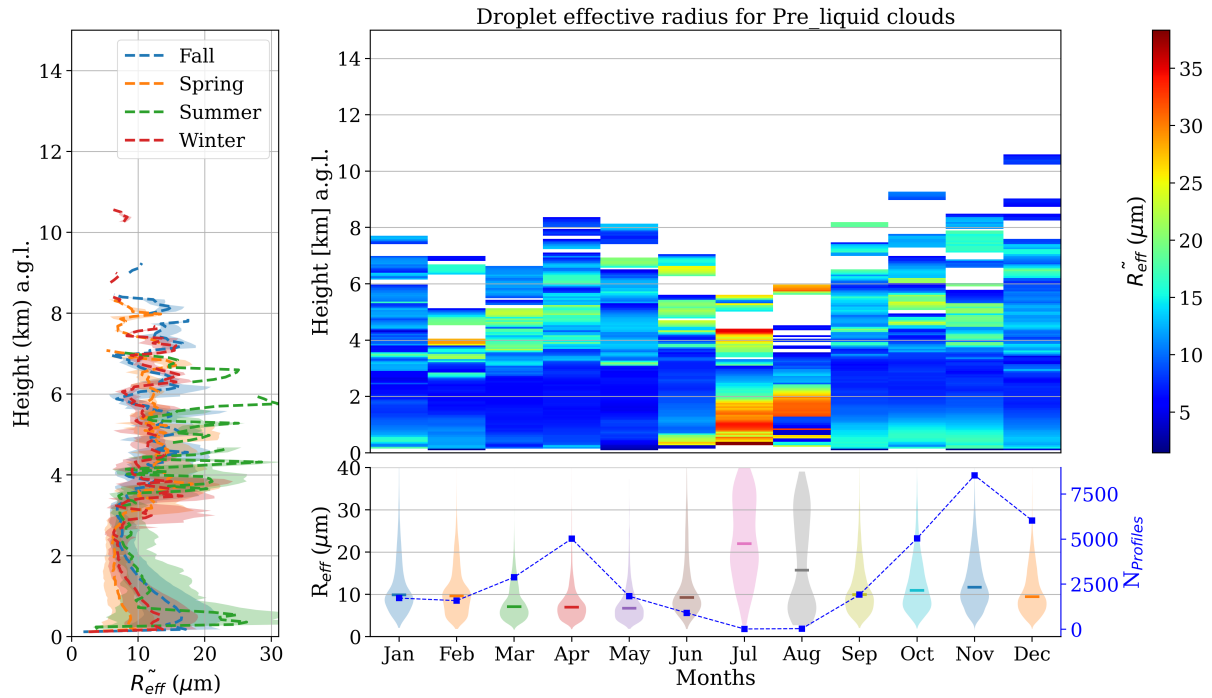


Figure 16. Same as figure 15, but for precipitable liquid clouds.

Concerning mixed-phase clouds (Figure 17), no seasonal differences were observed. Nonetheless, Left panel of Figure 17 shows that droplets of mixed-phase clouds are systematically larger than within liquid clouds, roughly twice the size. In addition, larger cloud droplets are found between 3 and 5 km of height. This is also observed by the monthly median profiles in the right-top panel, by the yellow and red tones. Same panel also indicates that greater droplets sizes are found in summer, despite the fact there are small amounts of LWP in this season as observed in Figure 9. Even though an evaluation of droplet size distribution on the right-bottom panel of Figure 17 shows an stable median around $10 \mu\text{m}$ through the months, droplets can vary between $13 - 22 \mu\text{m}$ in the neighbourhood of 4.5 km of height in summer. However, values are typically much smaller than this, as can be seen by the monthly p.d.f in the right-bottom panel. They show a mode below the median during the entire year, revealing that the most frequent droplet sizes are even smaller than $15 \mu\text{m}$.

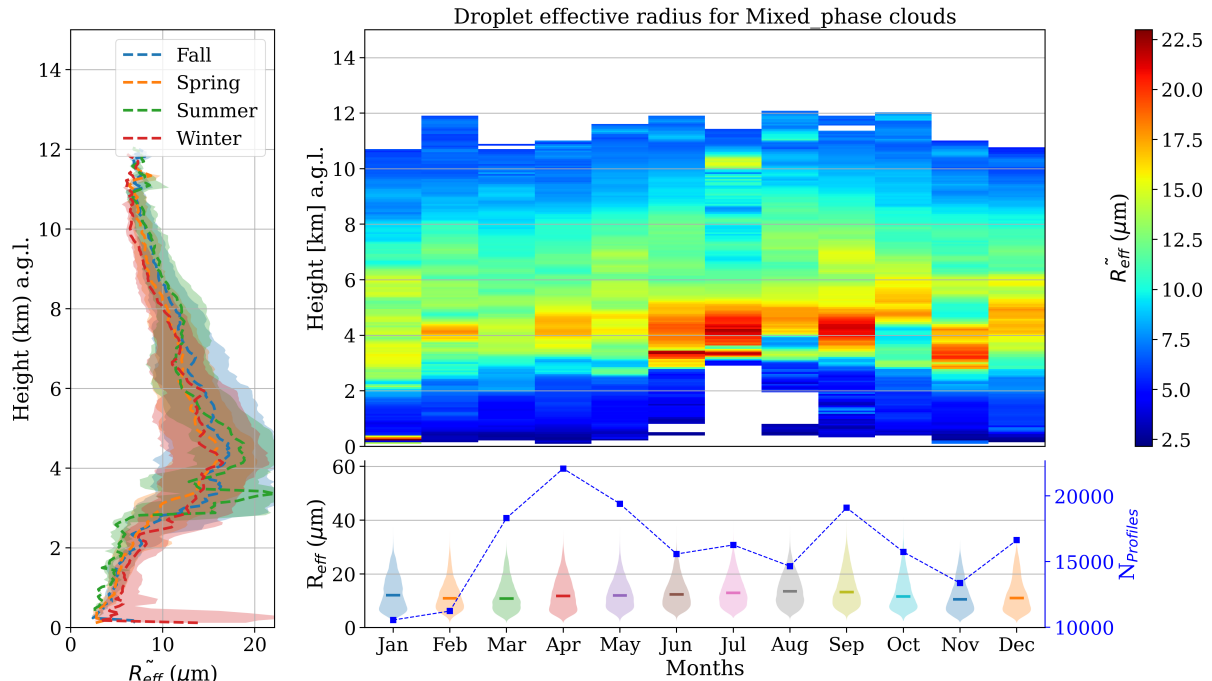


Figure 17. Same as figure 15, but for mixed-phase clouds.

Differently, precipitable mixed-phase clouds do not exhibit vertical variation of droplet size. Their median values were kept almost the same along the vertical by each median profile of Figure 18. Also, summer showed a slightly higher median profile of r_e than other seasons, roughly $10 \mu\text{m}$ greater than winter, between 3 and 6 km. Differences below this layer are quite challenging to analyze since these clouds have a cloud base about 3 km in summer, as already observed in Figure 6. Thus, besides these clouds having almost no occurrence in summer, there is almost no cloud base below this level. Therefore, a seasonal comparison of r_e should be done above this level, which still has larger r_e , but differences are less discrepant. An evaluation of the monthly distributions on the right-bottom panel also attribute less weight to these low clouds during summer. As a consequence, a difference around $5 \mu\text{m}$ was also observed between winter and summer medians. In summer, the median is varying around 20μ , and during the winter, the median is about $15 \mu\text{m}$. In addition, during summer it is possible to observe that the position of the mode, which is located below the median for the rest of the period, starts to move until stays above the median in August.

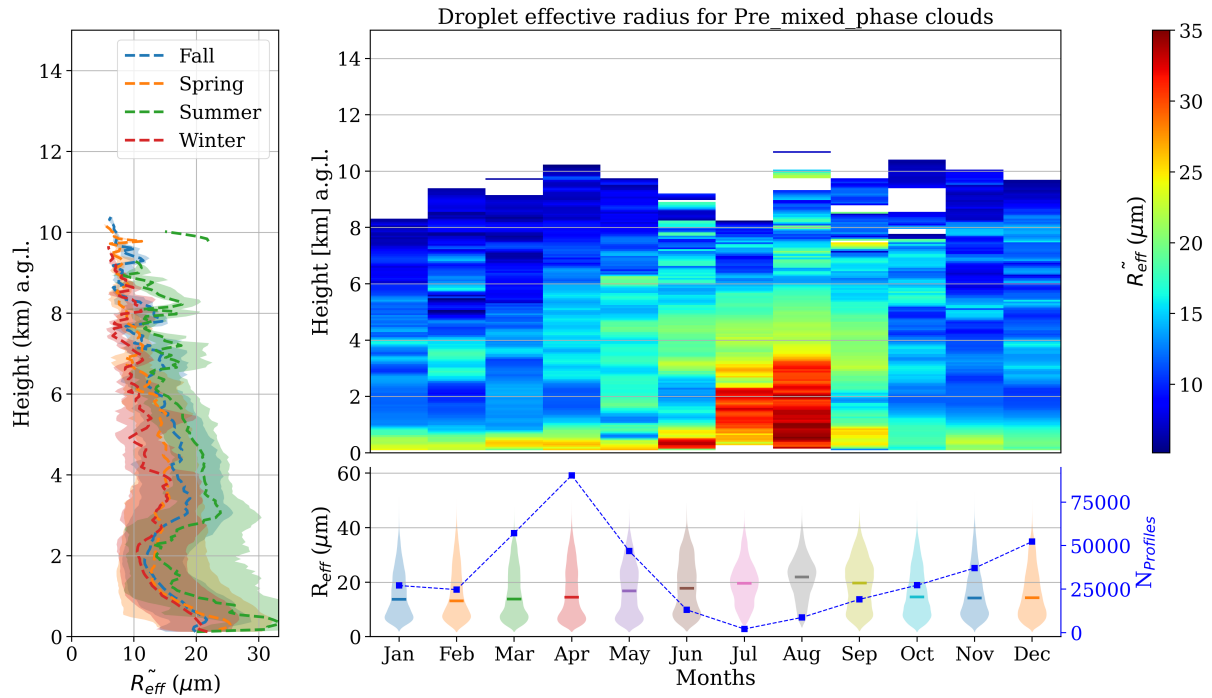


Figure 18. Same as figure 15, but for precipitable mixed-phase clouds.

In general, no strong seasonal differences were observed for liquid and precipitable liquid clouds. Liquid clouds have r_e varying around $5 \mu\text{m}$, while precipitable clouds achieved higher values due to the presence of drizzle droplets, varying between $10 \mu\text{m}$. Mixed-phase clouds showed consistently higher median r_e compared to liquid clouds, where larger values occurred around 3 and 5 km of height, especially during summer. Since just a few fraction of these clouds have larger r_e , and statistics with all monthly data showed only small variation of the month medians, which stayed around $15 \mu\text{m}$. In relation to precipitable mixed-phase clouds, larger differences observed in summer occurred due to the presence of a few low clouds, which were higher r_e . However, these clouds do not have enough occurrence to compare with the other months, which have considerably strong statistics.

1.8 Ice effective radius

Ice effective radius for ice clouds are depicted in Figure 19, and is organized as the previous droplet sizes analysis. Left panel shows a slight seasonal difference above 7 km, where summer assumed larger median r_e . Also, larger values of ice r_e is observed in summer by the yellow tone bellow 4 km in the right-top panel, indicating a radius about $60 \mu\text{m}$. However, this clouds have a median cloud base around 7 km in summer, in addition to a 25th percentile of 5 km. This shows that there is a negligible occurrence of ice clouds bellow 4 km. Therefore, r_e for these clouds are typically ranging between 30 and $50 \mu\text{m}$, where lower values are more frequently. This can be also seen in the monthly p.d.fs on the right-bottom panel. Throughout the observation period, the median remained close to $30 \mu\text{m}$, and increased in summer due to the small presence of low ice clouds.

During this season, median r_e are close to $50 \mu\text{m}$. It is worth to note the importance of analysing the median instead of the mean in these cases. Clearly, outliers close to surface, which does not represent a real situation for ice clouds will affect the average in May, August, September and October. The monthly distribution of r_e also shows the presence of a mode bellow the median for almost all season, except for April and summer. As a result, common r_e can be even smaller than $30 \mu\text{m}$

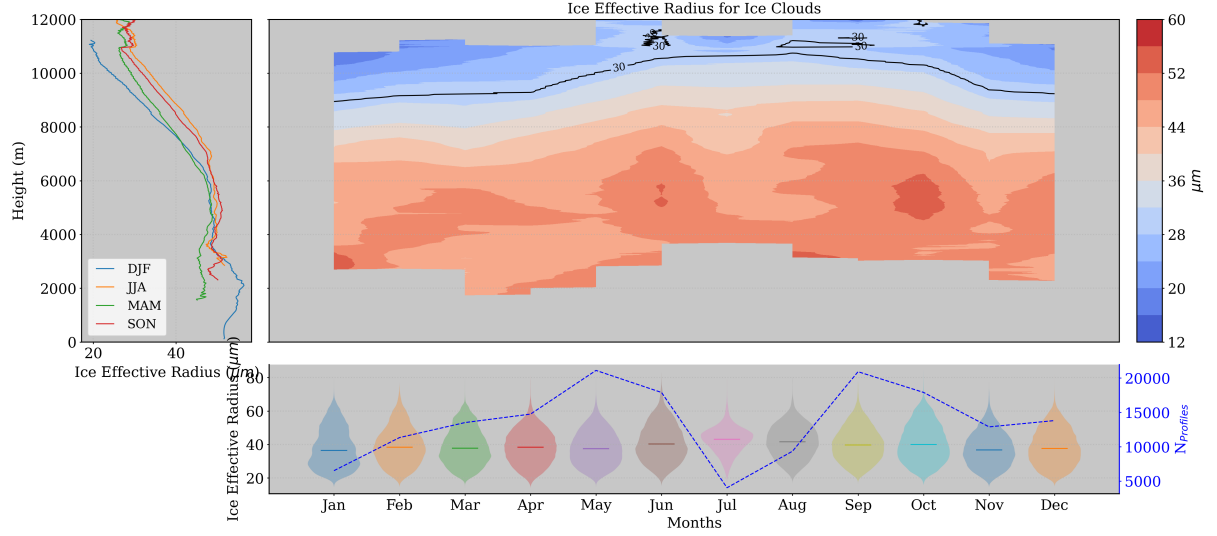


Figure 19. Seasonal and monthly statistics on ice effective radius (r_e) distribution for ice clouds. Left panel shows the seasonal median profiles of ice effective radius, where shaded areas denote the 25th and 75th percentile. Top-right panel shows the monthly profiles of the median r_e , while the bottom-panel displays the size distribution of r_e by months.

For mixed-phase clouds, the left panel of Figure 20 also shows a slight difference between profiles above 6 km. In summer and fall, median values are slightly larger than other seasons. The reason can be associated with the vertical location of clouds at each season, as cloud bases in summer and fall are around 5 km, and one kilometer lower in winter and spring. Hence, high mixed-phase clouds in summer and fall are exhibiting large r_e . Seasonal differences below 3 km are quite challenging to evaluate due to the negligible presence of these clouds. As a reference, the 25th percentile of cloud base height in summer is 4.6 km, denoting a small presence of clouds below this level. However, regarding the entire dataset, the distributions in the right-bottom panel presented slightly larger values in summer and fall, reaching at $50 \mu\text{m}$ in July, August and September. In general, median r_e are just a little smaller, with high variability. Different from the previous case, the median and the mode almost coincide for all the months.

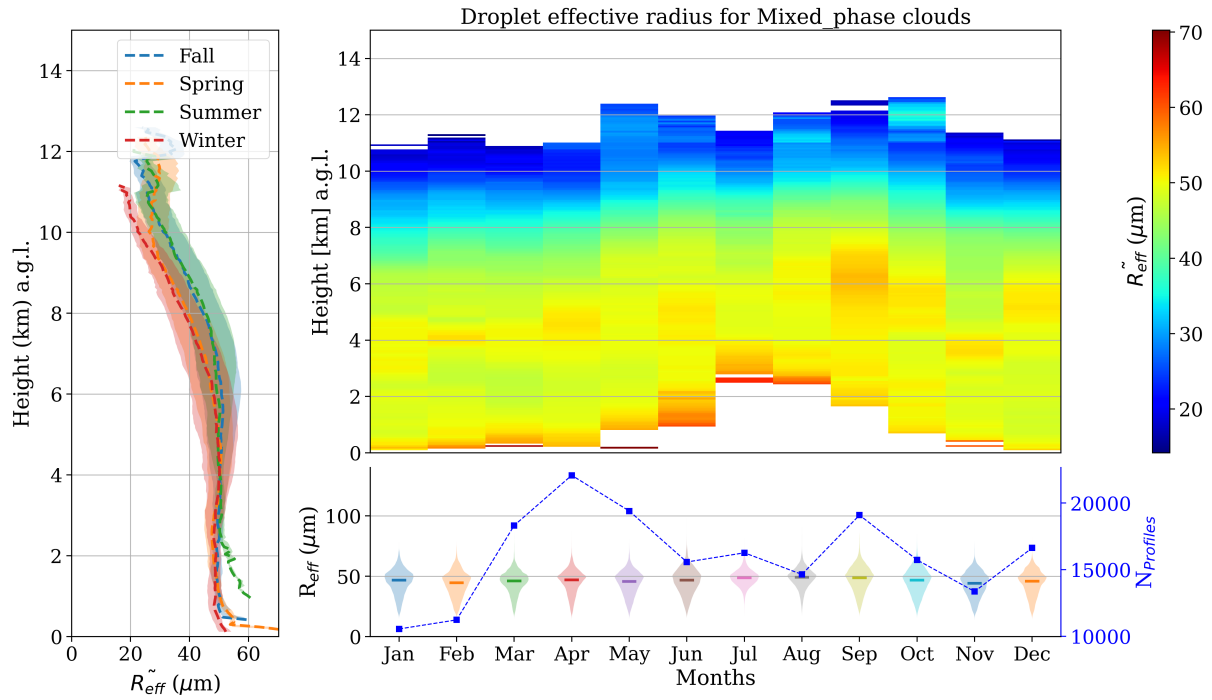


Figure 20. Same as figure 19, but for mixed-phase clouds.

Lastly, precipitable mixed-phase clouds experienced the greater seasonal difference between median profiles. Above five kilometers, summer has the largest median values, followed by fall and then winter and spring with the same values. Differences above 7 km cannot be analyzed since cloud tops are frequently not observed at this level, with the 75th percentile at 6 and 4.5 km in fall and spring, respectively. Probably, the differences observed above 12 km are due to a bad classification of cloud layers. Same reason can be causing values about 70 μm at 10 km observed in the right-top panel of Figure 20, in July. In addition, this month has a poor number of profiles, denoting just minor cases that can be contributing to these high values. In general, the statistical median of the monthly p.d.fs is stable in 50 μm , with a narrower distribution.

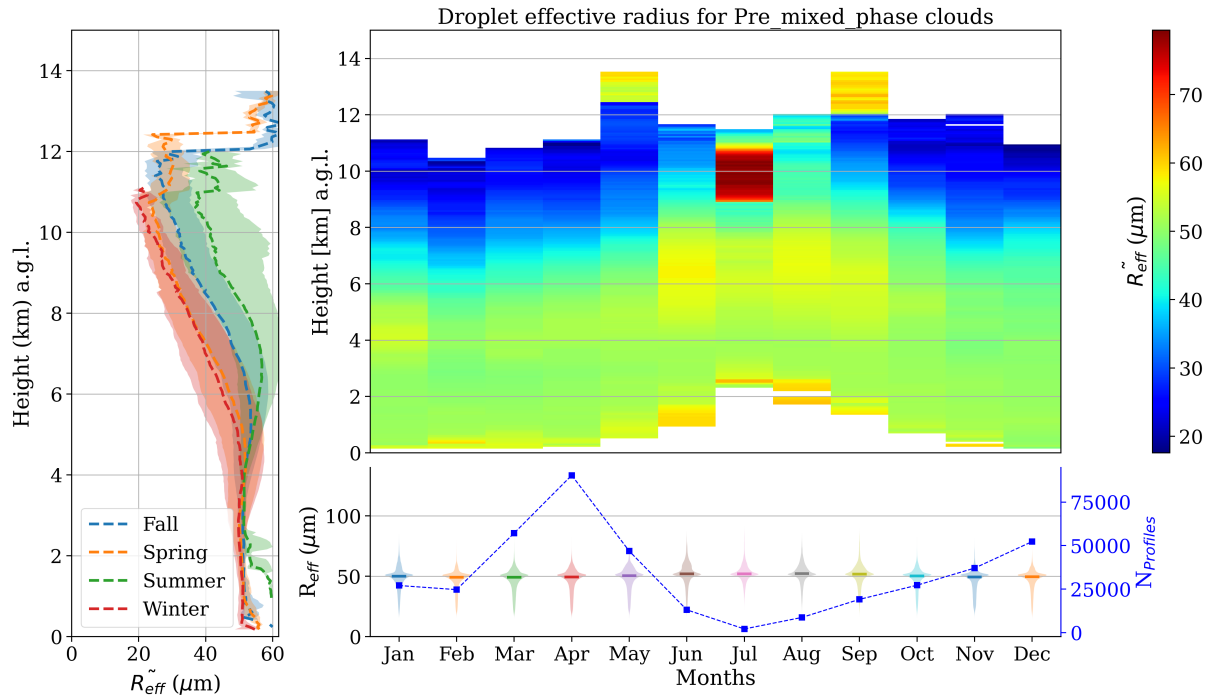


Figure 21. Same as figure 19, but for precipitable mixed-phase clouds.

In summary, ice clouds have median r_e ranging between 30 and 50 μm, where values can be larger at more elevated levels. The second is commonly found during summer, and the first during the rest of the period. For mixed-phase and precipitable mixed-phase clouds, the median r_e are about 50 μm. In fact, mixed-phase clouds have slightly smaller values than 50 μm excluding summer, and a broader distribution. However, values are quite similar with the precipitable case.

Acknowledgements. TEXT

References

- Alexandre Rios-Entenza, Rios-Entenza, A., Pedro M. M. Soares, Soares, P. M. M., Ricardo M. Trigo, Trigo, R. M., Rita M. Cardoso, 325 Cardoso, R. M., Gonzalo Miguez-Macho, and Miguez-Macho, G.: Moisture Recycling in the Iberian Peninsula from a Regional Climate Simulation: Spatiotemporal Analysis and Impact on the Precipitation Regime, 119, 5895–5912, <https://doi.org/10.1002/2013jd021274>.
- Shahi, N. K., Polcher, J., Bastin, S., Pennel, R., and Fita, L.: Assessment of the Spatio-Temporal Variability of the Added Value on Precipitation of Convection-Permitting Simulation over the Iberian Peninsula Using the RegIPSL Regional Earth System Model, 59, 471–498, <https://doi.org/10.1007/s00382-022-06138-y>.